

✓ techExposed Lab #001: How the Internet Works

Lab Type: Idea

Estimated Time: 30–45 mins

Skill Level: Beginner

```
# Let's begin by printing your name to personalize the notebook
your_name = "Sudesh"
print(f"Welcome to the lab, {your_name}!")
```

```
Welcome to the lab, Sudesh!
```

STAR Method Lab Prompt

Situation

The technical system I am breaking down is DNS (Domain Name System). DNS is an important part of the internet, but it can be difficult for new developers and students to understand because most of its work happens automatically in the background. When someone types a website name such as youtube.com into a browser, they may not realize that the computer needs to find the website's IP address before it can connect to the correct server. DNS solves this problem by translating easy-to-remember domain names into IP addresses that computers can use. Understanding DNS is important because it is one of the first steps that happens when a user tries to access a website.

Task

The goal is to explain how DNS translates a human-readable domain name into an IP address and allows a browser to find and connect to the correct server. The explanation should show the basic steps of a DNS lookup, including how a computer sends a request to a DNS resolver, how the resolver finds the correct DNS information, and how the IP address is returned to the computer. The goal is to make the process simple enough for a new developer or student to understand without needing advanced networking knowledge.

Action

I will use Gemini, Markdown, and visual explanations to break the DNS process into simple steps. I will use Markdown headings and bullet points to organize the information and explain each part of the process in plain language. I will also use a simple visual flow such as youtube.com → DNS resolver → IP address → web server to show how the different components communicate. This visual approach will make it easier to understand what happens after a user enters a website address into a browser.

Expected Result

The expected result is an understandable explanation of DNS that allows a new developer or student to understand what DNS does, how a DNS lookup works, and why DNS is important for

accessing websites. By the end, the reader should understand that DNS acts like a directory for the internet by connecting human-readable domain names to IP addresses. The explanation should also give the reader a basic understanding of what happens behind the scenes when they enter a website address into their browser.

Your Assignment

Use this section to complete your deliverable:

(Example Format)

- **Technology Explored**: Internet stack (DNS → IP → HTTP → Hosting)
- **STAR Output**:
 - S: Many new developers and students do not understand how a domain name such as `youtube.com` is connected to the correct server because DNS works behind the scenes.
 - T: Explain how DNS translates a human-readable domain name into an IP address and helps a browser find the correct server.
 - A: Used Gemini, Markdown, and a simple visual flow to break down the DNS lookup process into easy-to-understand steps.
 - R: Created a clear explanation showing how `youtube.com` is translated into an IP address, allowing a new developer or student to understand the basic purpose and process of DNS.